

README.md for Car Damage Classification Project

Car Damage Classification using Custom PyTorch CNN

This repository contains a PyTorch implementation for classifying car damages into 4 categories. The model dynamically splits the raw dataset into training validation, and testing subsets, trains a custom 3-layer Convolutional Neural Network (CNN), saves the best-performing weights, and evaluates the final .accuracy

Project Overview

- Objective: Automatically classify vehicle damage to assist insurance claims and smart estimation systems
 - Architecture: Custom Convolutional Neural Network (CNN) built with PyTorch
 - Input Size: 224×224 pixels (RGB)
 - Classes (4-categories):
 - Scratch (خراشیدگی)
 - Dent (فرورفتگی)
 - Broken Glass (شکستگی شیشه)
 - Intact / Undamaged (سالم)
-

Project Structure

```
model.py      # Custom CNN architecture (CarDamageCNN) —|  
               /src —|  
               /car-damage-classification
```

```
dict.py      # Inference script for classifying a single image —┐ |
# Raw dataset grouped by classes (local only - gitignored) —┐ |
plit/      # Formatted split folders (train/val/test - gitignored) —┐ |
# Main pipeline: splits dataset, trains, validates & tests —┐ |
              requirements.txt      # Python dependencies —┐ |
training_curves.png # Generated evaluation metrics plot —┐ |
              README.md             # Project documentation —┐ |
```

Installation & Setup

```
-bash git clone https://github.com/YOUR :Clone the Repository .1
-USERNAME/car-damage-classification.git cd car-damage
classification
```

```
Make sure you have python 3.8+ installed. You can :Install Dependencies .2
bash pip install -r :install all requirements via
requirements.txt
```

```
.folder dataset Place your raw images inside the :Data Organization .3
—┐ /text dataset/ |— scratch :Keep them grouped by class
/dent/ |— broken_glass/ |— intact
```

Training and Evaluation

,To split the dataset, start the training process, and save the best model weights
:run

```
python main.py
```

:Key Training Features

- .split 80%/10%/10% :Train/Val/Test Split •
- Automatically tracks the lowest validation loss to save :Early Saving •
- . best_car_damage_model.pth

visual showing Loss and `training_curves.png` A :Output Plots
.Accuracy histories is saved automatically at the end of execution

Making Predictions

:To run inference on a new, unseen image

```
from src.predict import predict_image

image_path = "path/to/test_car.jpg"
prediction = predict_image(image_path)
print(f"Detected Damage Type: {prediction}")
```

Results & Visualization

Add your training curve image and sample predictions here once you run the)
(!project on your full dataset

Placeholder